

Practical Week 10

Session 10

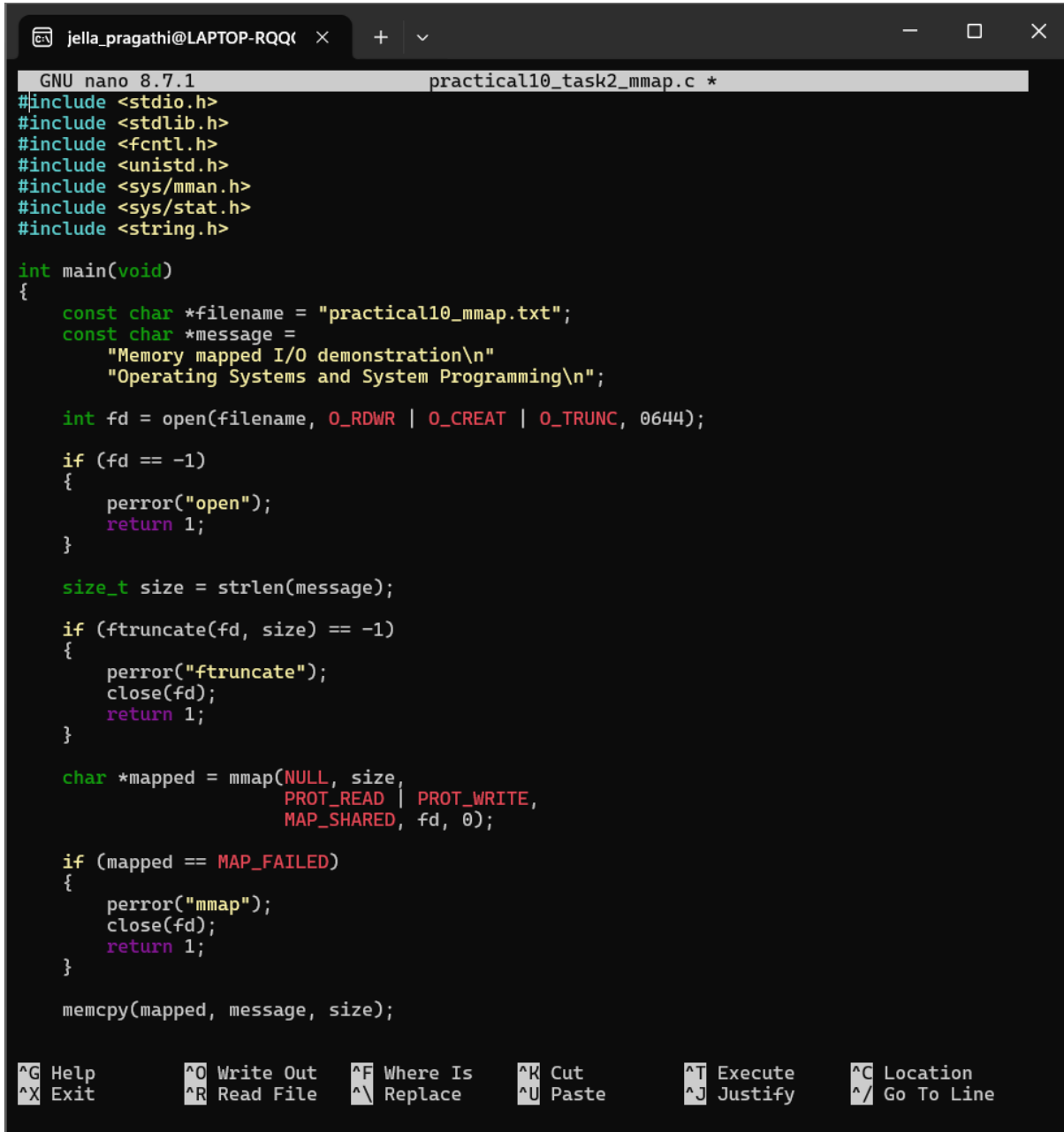
Task 1

Investigate inode structures using commands such as `ls -li`, `stat`, and `find`. Create hard links and symbolic links and explain their impact on inode allocation.

```
jella_pragathi@LAPTOP-RQQGT555:~$ cd ~
jella_pragathi@LAPTOP-RQQGT555:~$ echo "Operating Systems inode demonstration" > practical10_original.txt
jella_pragathi@LAPTOP-RQQGT555:~$ ls -li practical10_original.txt
40892 -rw-rw-r-- 1 jella_pragathi jella_pragathi 38 Sep 28 05:36 practical10_original.txt
jella_pragathi@LAPTOP-RQQGT555:~$ stat practical10_original.txt
  File: practical10_original.txt
  size: 38          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 40892         Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1001/jella_pragathi)   Gid: ( 1001/jella_pragathi)
Access: 2026-09-28 05:36:27.480992520 +0000
Modify: 2026-09-28 05:36:27.480992520 +0000
Change: 2026-09-28 05:36:27.480992520 +0000
 Birth: 2026-09-28 05:36:27.480992520 +0000
jella_pragathi@LAPTOP-RQQGT555:~$ find ~ -inum 40822
jella_pragathi@LAPTOP-RQQGT555:~$ ln practical10_original.txt practical10_hardlink.txt
jella_pragathi@LAPTOP-RQQGT555:~$ ls -li practical10_original.txt practical10_hardlink.txt
40892 -rw-rw-r-- 2 jella_pragathi jella_pragathi 38 Sep 28 05:36 practical10_hardlink.txt
40892 -rw-rw-r-- 2 jella_pragathi jella_pragathi 38 Sep 28 05:36 practical10_original.txt
jella_pragathi@LAPTOP-RQQGT555:~$ ln -s practical10_original.txt practical10_symlink.txt
jella_pragathi@LAPTOP-RQQGT555:~$ ls -li practical10_original.txt practical10_hardlink.txt practical10_symlink.txt
40892 -rw-rw-r-- 2 jella_pragathi jella_pragathi 38 Sep 28 05:36 practical10_hardlink.txt
40892 -rw-rw-r-- 2 jella_pragathi jella_pragathi 38 Sep 28 05:36 practical10_original.txt
40893 lrwxrwxrwx 1 jella_pragathi jella_pragathi 24 Sep 28 05:38 practical10_symlink.txt -> practical10_original.txt
jella_pragathi@LAPTOP-RQQGT555:~$ cat practical10_original.txt
cat practical10_hardlink.txt
cat practical10_symlink.txt
Operating Systems inode demonstration
Operating Systems inode demonstration
Operating Systems inode demonstration
jella_pragathi@LAPTOP-RQQGT555:~$ |
```

Task 2

Design a program using `mmap()` to perform file reading and writing. Compare memory-mapped I/O with traditional read/write operations in terms of efficiency and implementation complexity.



```
GNU nano 8.7.1 practical10_task2_mmap.c *
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <string.h>

int main(void)
{
    const char *filename = "practical10_mmap.txt";
    const char *message =
        "Memory mapped I/O demonstration\n"
        "Operating Systems and System Programming\n";

    int fd = open(filename, O_RDWR | O_CREAT | O_TRUNC, 0644);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    size_t size = strlen(message);

    if (ftruncate(fd, size) == -1)
    {
        perror("ftruncate");
        close(fd);
        return 1;
    }

    char *mapped = mmap(NULL, size,
                        PROT_READ | PROT_WRITE,
                        MAP_SHARED, fd, 0);

    if (mapped == MAP_FAILED)
    {
        perror("mmap");
        close(fd);
        return 1;
    }

    memcpy(mapped, message, size);
}
```

Help	Write Out	Where Is	Cut	Execute	Location
Exit	Read File	Replace	Paste	Justify	Go To Line

```

    if (msync(mapped, size, MS_SYNC) == -1)
    {
        perror("msync");
        munmap(mapped, size);
        close(fd);
        return 1;
    }

    printf("Data written using mmap():\n");
    printf("%.5s", (int)size, mapped);

    if (munmap(mapped, size) == -1)
    {
        perror("munmap");
        close(fd);
        return 1;
    }

    close(fd);

    printf("\nFile successfully written using memory-mapped I/O.\n");

    return 0;
}

```

^{^G} Help ^{^O} Write Out ^{^F} Where Is ^{^K} Cut ^{^T} Execute ^{^C} Location
^{^X} Exit ^{^R} Read File ^{^N} Replace ^{^U} Paste ^{^J} Justify ^{^_} Go To Line

```

See 'snap info <snapname>' for additional versions.
jella_pragathi@LAPTOP-RQQT555:~$ nano practical10_task2_mmap.c
jella_pragathi@LAPTOP-RQQT555:~$ gcc -Wall -Wextra practical10_task2_mmap.c -o practical10_task2_mmap
jella_pragathi@LAPTOP-RQQT555:~$ ./practical10_task2_mmap
Data written using mmap():
Memory mapped I/O demonstration
Operating Systems and System Programming

File successfully written using memory-mapped I/O.
jella_pragathi@LAPTOP-RQQT555:~$ cat practical10_mmap.txt
Memory mapped I/O demonstration
Operating Systems and System Programming
jella_pragathi@LAPTOP-RQQT555:~$ ls -l practical10_mmap.txt
-rw-r--r-- 1 jella_pragathi jella_pragathi 73 Sep 28 05:42 practical10_mmap.txt
jella_pragathi@LAPTOP-RQQT555:~$

```